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Thermodynamics and Statistical Physics

Spring 2026

Homework 6

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Student Information

Name:	Xu Chen
Student ID:	523110910017
Email:	xuchen2005@sjtu.edu.cn

Show all essential steps in your derivations.

Negative temperature for a system of N Ising spins $\star\star\star\star$

Consider an isolated system of $N \gg 1$ noninteracting Ising spins in equilibrium. Each spin is described by a variable $q_i \in \{0, 1\}$, with energy $\epsilon\, q_i$. The total energy is

$$E(x) = \epsilon \sum_i q_i. \tag{1}$$

Consider a Boltzmann state where M spins are excited, so that the total energy is

$$E = M\epsilon,$$

and all compatible microstates are equally probable.

(a) Compute the number of microstates $\Omega(E)$ and show that

$$\Omega(E) = \binom{N}{M}.$$

Compute the Boltzmann entropy $S(E) = \log \Omega(E)$ using Stirling's approximation, and express it in terms of $m = M/N$. Compute the temperature using

$$\frac{1}{T} = \beta = \frac{dS}{dE}.$$

Show that $T < 0$ if $m > 1/2$.

(b) With m fixed, consider a single spin as a subsystem and compute its marginal distribution $p(x)$. Show that it can be written in the canonical form

$$p(x) = \frac{e^{-\beta\epsilon\, x}}{z}, \qquad z = 1 + e^{-\beta\epsilon},$$

with β given in part (a).

(c) Compute the heat capacity $C = \frac{dE}{dT}$ and show that $C > 0$ even when $T < 0$.

(d) According to the second law, a thermodynamic state is stable if it maximizes entropy under the relevant constraints. Is a Boltzmann state with $T < 0$ for this N-spin system stable?

(e) Now place the N-spin system in weak thermal contact with a bath at positive temperature $T' > 0$. Construct an infinitesimal energy exchange and compute the change in total entropy. What is the direction of energy flow? Does it mean that a system with $T < 0$ is "hotter" than a system with $T > 0$?

(f) Construct a reversible engine operating between two spin reservoirs, one at temperature $T_1 < 0$ and the other at temperature $T_2 > 0$. (Here reversible means that the total entropy of the two reservoirs is conserved.) Compute the efficiency of this engine and show that it is formally still given by the Carnot formula, and is larger than 1. Does this violate the first or the second law of thermodynamics?

(g) Now place the N-spin system in weak contact with an infinite heat bath at positive temperature $T > 0$, so that the system is described by the canonical ensemble

$$P(x) = \frac{e^{-\beta E(x)}}{Z}, \qquad \beta = \frac{1}{T}.$$

(i) Show that the probability distribution of $M = \sum_i q_i$ is

$$P(M) = \frac{\binom{N}{M} e^{-\beta\epsilon M}}{(1 + e^{-\beta\epsilon})^N}.$$

(ii) Using Stirling's approximation, show that for large N this distribution satisfies a large deviation principle of the form

$$P(M) \asymp e^{-N I_\beta(m)},$$

and determine the rate function $I_\beta(m)$.

(iii) Recall from part (a) that a microcanonical state has negative temperature when $m > 1/2$. Estimate the probability that the system visits the negative-temperature region,

$$\text{Prob}(m > 1/2).$$

Solution

(a) The system consists of N spins, and the total energy is $E = M\epsilon$, meaning exactly M spins are in the excited state ($q_i = 1$) and the remaining $N - M$ spins are in the ground state ($q_i = 0$). The number of ways to choose M excited spins out of N is simply the binomial coefficient:

$$\Omega(E) = \binom{N}{M} = \frac{N!}{M!(N-M)!}.$$

Using Stirling's approximation $\log N! \approx N \log N - N$, the Boltzmann entropy is:

$$\begin{aligned} S(E) &= \log \Omega(E) \\ &\approx (N \log N - N) - (M \log M - M) - ((N-M)\log(N-M) - (N-M)) \\ &= N \log N - M \log M - (N-M)\log(N-M). \end{aligned}$$

Let $m = M/N$. Substituting $M = Nm$ and $N - M = N(1-m)$:

$$\begin{aligned} S(E) &= N \log N - Nm \log(Nm) - N(1-m)\log(N(1-m)) \\ &= -N\left[m \log m + (1-m)\log(1-m)\right]. \end{aligned}$$

The thermodynamic temperature is defined via $\beta = \frac{1}{T} = \frac{dS}{dE}$. Using the chain rule

$\frac{dS}{dE} = \frac{dS}{dm}\frac{dm}{dE} = \frac{dS}{dm}\frac{1}{N\epsilon}$:

$$\begin{aligned}\beta &= \frac{1}{N\epsilon}\frac{d}{dm}\left(-N\left[m\log m + (1-m)\log(1-m)\right]\right)\\ &= -\frac{1}{\epsilon}\left[\log m + 1 - \log(1-m) - 1\right]\\ &= \frac{1}{\epsilon}\log\left(\frac{1-m}{m}\right).\end{aligned}$$

If $m > 1/2$, then $\frac{1-m}{m} < 1$, which implies $\log\left(\frac{1-m}{m}\right) < 0$. Therefore, $\beta < 0$, which corresponds to $T < 0$.

(b) Consider a single spin. In the microcanonical ensemble with M excitations, all $\binom{N}{M}$ microstates are equally likely. The number of microstates where a specific spin $q_1 = 1$ is $\binom{N-1}{M-1}$ (choosing $M-1$ excitations from the remaining $N-1$ spins). The probability is:

$$p(1) = \frac{\binom{N-1}{M-1}}{\binom{N}{M}} = \frac{M}{N} = m.$$

Similarly, the probability that $q_1 = 0$ is:

$$p(0) = \frac{\binom{N-1}{M}}{\binom{N}{M}} = \frac{N-M}{N} = 1-m.$$

We want to write this in the canonical form $p(x) = \frac{e^{-\beta\epsilon x}}{z}$ for $x \in \{0,1\}$. From part (a), we know $\beta\epsilon = \log\left(\frac{1-m}{m}\right) \implies e^{-\beta\epsilon} = \frac{m}{1-m}$. Evaluate the canonical probabilities:

$$\begin{aligned}p(1) &= \frac{e^{-\beta\epsilon}}{z} = \frac{\frac{m}{1-m}}{z}\\ p(0) &= \frac{e^{0}}{z} = \frac{1}{z}\end{aligned}$$

Since $p(0) = 1-m$, we immediately identify $z = \frac{1}{1-m}$. Let's verify $z = 1 + e^{-\beta\epsilon}$:

$$1 + e^{-\beta\epsilon} = 1 + \frac{m}{1-m} = \frac{1}{1-m} = z.$$

Thus, $p(x) = \frac{e^{-\beta\epsilon x}}{z}$ correctly reproduces the marginal distribution.

(c) From $\beta = \frac{1}{\epsilon}\log\left(\frac{1-m}{m}\right)$, we invert to find $m(\beta)$:

$$e^{\beta\epsilon} = \frac{1-m}{m} \implies m = \frac{1}{e^{\beta\epsilon}+1}.$$

The total energy is $E = N\epsilon m = \frac{N\epsilon}{e^{\beta\epsilon}+1}$. The heat capacity is $C = \frac{dE}{dT} = \frac{dE}{d\beta}\frac{d\beta}{dT} = \frac{dE}{d\beta}\left(-\frac{1}{T^2}\right)$. First, evaluate $\frac{dE}{d\beta}$:

$$\frac{dE}{d\beta} = N\epsilon\frac{d}{d\beta}(e^{\beta\epsilon}+1)^{-1} = -N\epsilon^2\frac{e^{\beta\epsilon}}{(e^{\beta\epsilon}+1)^2}.$$

Now multiply by $-1/T^2$:

$$C = \frac{N\epsilon^2}{T^2}\frac{e^{\beta\epsilon}}{(e^{\beta\epsilon}+1)^2}.$$

Since all terms in this expression $(N, \epsilon^2, e^{\beta\epsilon}, (e^{\beta\epsilon}+1)^2)$ are strictly positive, and $T^2 > 0$ regardless of whether T is positive or negative, we conclude that $C > 0$ even when $T < 0$.

(d) According to the second law of thermodynamics, a state is stable if it uniquely maximizes the entropy under the given constraints. Mathematically, this implies the entropy function $S(E)$ must be concave, meaning $\frac{d^2S}{dE^2} < 0$. We have:

$$\frac{d^2S}{dE^2} = \frac{d}{dE}\left(\frac{dS}{dE}\right) = \frac{d}{dE}\left(\frac{1}{T}\right) = -\frac{1}{T^2}\frac{dT}{dE} = -\frac{1}{T^2C}.$$

Because $C > 0$ (proven in part c), it follows that $\frac{d^2S}{dE^2} < 0$. Thus, $S(E)$ is strictly concave everywhere. The Boltzmann state with $T < 0$ corresponds to a legitimate entropy maximum and is therefore absolutely stable.

(e) Consider a weak thermal contact with energy exchange dE. Conservation of energy dictates $dE_{\text{sys}} + dE_{\text{bath}} = 0$, so $dE_{\text{bath}} = -dE_{\text{sys}}$. The change in total entropy is:

$$dS_{\text{tot}} = dS_{\text{sys}} + dS_{\text{bath}} = \frac{dE_{\text{sys}}}{T} + \frac{dE_{\text{bath}}}{T'} = \left(\frac{1}{T} - \frac{1}{T'}\right) dE_{\text{sys}}.$$

According to the second law, isolated spontaneous processes must have $dS_{\text{tot}} > 0$. Given $T < 0$ and $T' > 0$, we have $\frac{1}{T} < 0$ and $\frac{1}{T'} > 0$. Therefore, the bracketed term $\left(\frac{1}{T} - \frac{1}{T'}\right)$ is strictly negative. To ensure $dS_{\text{tot}} > 0$, we must have $dE_{\text{sys}} < 0$.

Direction of energy flow: Energy flows *from* the negative-temperature system *to* the positive-temperature bath.

Conclusion: Yes, a system with $T < 0$ is thermodynamically "hotter" than any system with $T > 0$ because heat spontaneously flows from the negative temperature system to the positive temperature one.

(f) Let the engine extract heat Q_1 from the $T_1 < 0$ reservoir and deposit heat Q_2 into the $T_2 > 0$ reservoir. The work done is $W = Q_1 - Q_2$. (We assume $Q_1 > 0$). For a reversible engine, total entropy is conserved:

$$\Delta S_{\text{tot}} = -\frac{Q_1}{T_1} + \frac{Q_2}{T_2} = 0 \implies Q_2 = Q_1\frac{T_2}{T_1}.$$

Since $T_1 < 0$ and $T_2 > 0$, we have $T_2/T_1 < 0$. Thus, $Q_2 < 0$. Physically, this means the engine does not deposit heat into the second reservoir; instead, it *extracts* heat from the T_2 reservoir as well (magnitude $|Q_2| = -Q_2$). The efficiency is defined as the work output divided by the heat taken from the "hot" reservoir (T_1):

$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{T_2}{T_1}.$$

Because $\frac{T_2}{T_1} < 0$, the term $-\frac{T_2}{T_1}$ is positive, giving $\eta > 1$.

First Law: Not violated. Energy is conserved as $W = Q_1 + |Q_2|$; the work done is simply the sum of the heat extracted from both reservoirs.

Second Law: Not violated. A negative temperature reservoir *increases* its entropy when it loses heat ($dS = dQ/T$, if $dQ < 0$ and $T < 0$, then $dS > 0$). Thus, extracting heat from a negative temperature reservoir is an entropy-increasing process, which makes it permissible to perfectly convert that heat (plus heat from another source) into work without violating the principle of entropy maximization.

(g) (i) The canonical partition function is $Z = \sum_x e^{-\beta E(x)} = \sum_x \prod_i e^{-\beta\epsilon q_i} = (1 + e^{-\beta\epsilon})^N$. The probability of observing macroscopic state M is the sum over all $\binom{N}{M}$ microstates that have energy $M\epsilon$:

$$P(M) = \frac{\sum_{x:E(x)=M\epsilon} e^{-\beta E(x)}}{Z} = \frac{\binom{N}{M} e^{-\beta\epsilon M}}{(1+e^{-\beta\epsilon})^N}.$$

(ii) Using Stirling's approximation on $\binom{N}{M}$ as derived in part (a), we have $\binom{N}{M} \approx e^{-N[m\log m + (1-m)\log(1-m)]}$, where $m = M/N$. Substituting this into $P(M)$:

$$\begin{aligned} P(M) &\approx \exp\left(-N[m\log m + (1-m)\log(1-m)]\right) \frac{e^{-\beta\epsilon N m}}{(1+e^{-\beta\epsilon})^N} \\ &= \exp\left(-N\left[m\log m + (1-m)\log(1-m) + \beta\epsilon m + \log(1+e^{-\beta\epsilon})\right]\right). \end{aligned}$$

This is in the large deviation form $P(M) \asymp e^{-NI_\beta(m)}$, where the rate function is:

$$I_\beta(m) = m\log m + (1-m)\log(1-m) + \beta\epsilon m + \log(1+e^{-\beta\epsilon}).$$

(iii) Since the bath temperature is $T > 0$, we have $\beta > 0$. The equilibrium mean value is $m_{eq} = \frac{1}{e^{\beta\epsilon}+1} < 1/2$. The large deviation rate function $I_\beta(m)$ reaches its absolute minimum (zero) at m_{eq}, and is monotonically increasing for $m > m_{eq}$. To estimate the probability of visiting the negative-temperature region ($m > 1/2$), the dominant contribution comes from the boundary of the region, $m = 1/2$. We evaluate the rate function at $m = 1/2$:

$$\begin{aligned} I_\beta(1/2) &= \frac{1}{2}\log\frac{1}{2} + \frac{1}{2}\log\frac{1}{2} + \frac{\beta\epsilon}{2} + \log(1+e^{-\beta\epsilon}) \\ &= -\log 2 + \frac{\beta\epsilon}{2} + \log(1+e^{-\beta\epsilon}) \\ &= \log\left(\frac{e^{\beta\epsilon/2} + e^{-\beta\epsilon/2}}{2}\right) = \log\cosh\left(\frac{\beta\epsilon}{2}\right). \end{aligned}$$

Therefore, the probability is exponentially suppressed as:

$$\text{Prob}(m > 1/2) \asymp e^{-NI_\beta(1/2)} = e^{-N\log\cosh(\beta\epsilon/2)} = \left[\cosh\left(\frac{\beta\epsilon}{2}\right)\right]^{-N}.$$

Three equivalent variational principles for the grand canonical ensemble $\star\star\star$

Consider an open system that can exchange both energy and particles with a large surrounding bath. The microstate of the system is denoted by x, with Hamiltonian $H(x)$ and particle-number function $N(x)$. Assume that the volume V is fixed.
Show that the grand canonical distribution

$$p_{\mathrm{GC}}(x) = \frac{1}{\Xi(T,V,\mu)} \exp\bigl[-\beta\bigl(H(x) - \mu N(x)\bigr)\bigr]$$

can be characterized equivalently by the following three variational principles.

1. **Maximum entropy with fixed average energy and particle number.**

   Consider all probability distributions $p(x)$ satisfying

   $$\int dx\, p(x) = 1, \qquad \int dx\, p(x)\, H(x) = \langle E\rangle, \qquad \int dx\, p(x)\, N(x) = \langle N\rangle.$$

   Show that the distribution that maximizes the Gibbs entropy

   $$S[p] = -\int dx\, p(x) \ln p(x)$$

   is precisely the grand canonical distribution.

2. **Maximum total entropy.**

   Now consider the system together with its bath as a larger isolated system with fixed total energy E_{tot} and fixed total particle number N_{tot}. If the system is in microstate x, then the bath has energy

   $$E' = E_{\mathrm{tot}} - H(x),$$

   and particle number

   $$N' = N_{\mathrm{tot}} - N(x).$$

   Assume that the bath is in equilibrium conditioned on (E', N'), with entropy

   $$S'(E', N').$$

   Show that the total entropy functional can be written as

   $$S_{\mathrm{tot}}[p] = \int dx\, p(x)\Bigl[-\ln p(x) + S'\bigl(E_{\mathrm{tot}} - H(x),\, N_{\mathrm{tot}} - N(x)\bigr)\Bigr].$$

   For a macroscopic bath, expand the bath entropy to first order around $(E_{\mathrm{tot}}, N_{\mathrm{tot}})$:

   $$S'(E_{\mathrm{tot}} - H,\, N_{\mathrm{tot}} - N) = S'(E_{\mathrm{tot}}, N_{\mathrm{tot}}) - \beta H + \beta\mu N + \cdots,$$

   where

   $$\beta = \left(\frac{\partial S'}{\partial E'}\right)_{N',V}, \qquad \mu = -T\left(\frac{\partial S'}{\partial N'}\right)_{E',V}.$$

   Show that maximizing $S_{\mathrm{tot}}[p]$ again leads to the grand canonical distribution.

3. **Minimization of the grand-potential functional.**

   Define the nonequilibrium grand-potential functional

$$\Phi[p] = \langle H\rangle - TS[p] - \mu\langle N\rangle.$$

   Show that

$$\Phi[p] = T\int dx\, p(x)\left[\ln p(x) + \beta\big(H(x) - \mu N(x)\big)\right].$$

   Explain why minimizing $\Phi[p]$ is equivalent to maximizing the total entropy in part (2), and conclude that its minimizer is again the grand canonical distribution.

4. Using the method of relative entropy, show that $p_{\mathrm{GC}}(x)$ is the global and unique maximizer of entropy subject to the constaints discussed above.

Solution

1. **Maximum entropy with fixed average energy and particle number.** We seek to maximize the functional $S[p] = -\int dx\, p(x)\ln p(x)$ subject to the constraints:

$$\int dx\, p(x) = 1, \qquad \int dx\, p(x)H(x) = \langle E\rangle, \qquad \int dx\, p(x)N(x) = \langle N\rangle.$$

   We use the method of Lagrange multipliers, constructing the unconstrained functional $\mathcal{L}[p]$:

$$\begin{aligned}\mathcal{L}[p] = &-\int dx\, p(x)\ln p(x) - \lambda_0\left(\int dx\, p(x) - 1\right)\\ &-\lambda_1\left(\int dx\, p(x)H(x) - \langle E\rangle\right)\\ &-\lambda_2\left(\int dx\, p(x)N(x) - \langle N\rangle\right).\end{aligned}$$

   Taking the functional derivative with respect to $p(x)$ and setting it to zero:

$$\frac{\delta\mathcal{L}}{\delta p(x)} = -\ln p(x) - 1 - \lambda_0 - \lambda_1 H(x) - \lambda_2 N(x) = 0.$$

   Solving for $p(x)$ gives:

$$p(x) = \exp\left(-(1+\lambda_0) - \lambda_1 H(x) - \lambda_2 N(x)\right).$$

   To match standard thermodynamic notation, we identify the multipliers as $\lambda_1 = \beta = 1/T$ and $\lambda_2 = -\beta\mu$. The normalization constant is extracted via $1 + \lambda_0 = \ln\Xi(T, V, \mu)$. Substituting these identifications yields:

$$p(x) = \frac{1}{\Xi(T,V,\mu)}\exp\left[-\beta\big(H(x) - \mu N(x)\big)\right] \equiv p_{\mathrm{GC}}(x).$$

2. **Maximum total entropy.** The total entropy functional is:

$$S_{\mathrm{tot}}[p] = \int dx\, p(x)\Big[-\ln p(x) + S'\big(E_{\mathrm{tot}} - H(x),\, N_{\mathrm{tot}} - N(x)\big)\Big].$$

Using the provided first-order Taylor expansion for a macroscopic bath:

$$S'(E_{\text{tot}} - H(x),\ N_{\text{tot}} - N(x)) \approx S'(E_{\text{tot}}, N_{\text{tot}}) - \beta H(x) + \beta\mu N(x).$$

Substituting this into the functional:

$$S_{\text{tot}}[p] = \int dx\, p(x)\Big[-\ln p(x) - \beta\big(H(x) - \mu N(x)\big) + S'(E_{\text{tot}}, N_{\text{tot}})\Big].$$

Note that $S'(E_{\text{tot}}, N_{\text{tot}})$ is a constant independent of $p(x)$. To maximize $S_{\text{tot}}[p]$ subject to normalization $\int dx\, p(x) = 1$, we introduce a multiplier α:

$$\mathcal{L}[p] = S_{\text{tot}}[p] - \alpha\left(\int dx\, p(x) - 1\right).$$

Taking the functional derivative:

$$\frac{\delta\mathcal{L}}{\delta p(x)} = -\ln p(x) - 1 - \beta\big(H(x) - \mu N(x)\big) + S'(E_{\text{tot}}, N_{\text{tot}}) - \alpha = 0.$$

Solving for $p(x)$:

$$p(x) \propto \exp\big[-\beta\big(H(x) - \mu N(x)\big)\big].$$

Applying the normalization constraint exactly recovers the grand canonical distribution $p_{\text{GC}}(x)$.

3. **Minimization of the grand-potential functional.** The nonequilibrium grand-potential functional is defined as:

$$\Phi[p] = \langle H\rangle - TS[p] - \mu\langle N\rangle.$$

Substituting the expressions for the averages and $S[p]$:

$$\begin{aligned}\Phi[p] &= \int dx\, p(x)H(x) - T\left(-\int dx\, p(x)\ln p(x)\right) - \mu\int dx\, p(x)N(x)\\ &= \int dx\, p(x)\big[H(x) + T\ln p(x) - \mu N(x)\big].\end{aligned}$$

Factoring out T and using $\beta = 1/T$, we get:

$$\Phi[p] = T\int dx\, p(x)\left[\ln p(x) + \beta\big(H(x) - \mu N(x)\big)\right].$$

Equivalence to part (2): Comparing this expression to the expanded $S_{\text{tot}}[p]$ from part 2, we notice:

$$S_{\text{tot}}[p] = S'(E_{\text{tot}}, N_{\text{tot}}) - \frac{1}{T}\Phi[p].$$

Because $S'(E_{\text{tot}}, N_{\text{tot}})$ and $T > 0$ are constants, the variations satisfy $\delta S_{\text{tot}} = -\frac{1}{T}\delta\Phi$. Therefore, maximizing the total entropy $S_{\text{tot}}[p]$ is mathematically strictly equivalent to minimizing the grand-potential functional $\Phi[p]$. Since maximizing $S_{\text{tot}}[p]$ gave $p_{\text{GC}}(x)$, minimizing $\Phi[p]$ must uniquely yield the grand canonical distribution $p_{\text{GC}}(x)$ as well.

4. **Using the method of relative entropy.** The relative entropy (Kullback-Leibler divergence) between an arbitrary distribution $p(x)$ and the grand canonical distribution $p_{\mathrm{GC}}(x)$ is defined as:

$$D(p \parallel p_{\mathrm{GC}}) = \int dx\, p(x) \ln\left(\frac{p(x)}{p_{\mathrm{GC}}(x)}\right).$$

We know that $D(p \parallel p_{\mathrm{GC}}) \geq 0$ for all probability distributions p, with equality holding if and only if $p(x) = p_{\mathrm{GC}}(x)$ almost everywhere. Take the natural log of $p_{\mathrm{GC}}(x)$:

$$\ln p_{\mathrm{GC}}(x) = -\ln \Xi - \beta\big(H(x) - \mu N(x)\big).$$

Substitute this into the expression for $\Phi[p]$ derived in part (3):

$$\begin{aligned}
\Phi[p] &= T \int dx\, p(x) \left[\ln p(x) - \big(-\beta\big(H(x) - \mu N(x)\big)\big)\right] \\
&= T \int dx\, p(x) \left[\ln p(x) - (\ln p_{\mathrm{GC}}(x) + \ln \Xi)\right] \\
&= T \int dx\, p(x) \ln \frac{p(x)}{p_{\mathrm{GC}}(x)} - T \ln \Xi \int dx\, p(x) \\
&= T D(p \parallel p_{\mathrm{GC}}) - T \ln \Xi.
\end{aligned}$$

Since $T > 0$, the minimum of $\Phi[p]$ occurs exactly when the relative entropy $D(p \parallel p_{\mathrm{GC}})$ attains its absolute minimum of 0. Due to the properties of relative entropy, this minimum is uniquely achieved when $p(x) = p_{\mathrm{GC}}(x)$. Furthermore, since minimizing $\Phi[p]$ is equivalent to maximizing the system's constrained entropy (part 1) and the total system-bath entropy (part 2), this guarantees that $p_{\mathrm{GC}}(x)$ is the global and unique maximizer of entropy subject to the respective constraints.